

Date: 9-4-2025

Curl and Divergence

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These two operations that can be performed on vector fields and one produces a vector field whereas the other produces a scalar field.

Curl:-

If $F = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ is a vector field on $\mathbb{R}^3$ and partial derivatives of P, Q and R all exist,

then

$$\text{Curl } F = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

or

$$\text{Curl } F = \nabla \times F = \begin{array}{c} \downarrow \\ \text{cross} \\ \text{product} \end{array} \begin{array}{|c|c|c|} \hline \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \hline \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \hline P & Q & R \\ \hline \end{array}$$

Exp:- If $F(x, y, z) = xz\mathbf{i} + xyz\mathbf{j} - y^2\mathbf{k}$, Find curl F .

Solution:-

$$\text{Curl } F = \nabla \times F = \begin{array}{|c|c|c|} \hline \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \hline \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \hline xz & xyz & -y^2 \\ \hline \end{array}$$

$$= \mathbf{i} \left[\frac{\partial}{\partial y} (-y^2) - \frac{\partial}{\partial z} (xyz) \right] - \mathbf{j} \left[\frac{\partial}{\partial x} (-y^2) - \frac{\partial}{\partial z} (xz) \right] + \mathbf{k} \left[\frac{\partial}{\partial x} (xyz) - \frac{\partial}{\partial y} (xz) \right]$$

$$= \mathbf{i} [-2y - xy] - \mathbf{j} [0 - x] + \mathbf{k} [yz - 0]$$

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$$= (-2y - xy)i - (-x)j + (yz)k$$

$$= \boxed{-y(2+x)i + xj + yzk} \rightarrow \text{vector field}$$

$\Rightarrow$ curl F produce vector field.

$\Rightarrow$ If $\text{curl } F = 0$ then F is a conservative vector field.